Systems and Methods for Electromagnetic Coupling, Actuation, and Application of Scalar Refractive Fields (ψ)

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Cross-Reference to Related Applications

Pursuant to 35 U.S.C. §119(e), this application claims the benefit of the following U.S. provisional applications, the entirety of each being incorporated herein by reference:

- U.S. Provisional App. No. **63/865,279** (Confirmation No. 5785), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Gravitational and Signal Correction.*
- U.S. Provisional App. No. **63/865,283** (Confirmation No. 5905), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Energy Transmission.*
- U.S. Provisional specification filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Information Encoding, Transmission, and Computing.*
- Applicant's *Methods, Systems, and Apparatus for Artificial Generation, Control, and Application of Scalar Density Fields*, filed **Aug. 21, 2025**.

The foregoing applications are incorporated by reference in their entirety for all purposes, including definitions, figures, and embodiments. To the extent of any inconsistency, the present disclosure controls.

Field of the Invention

The disclosure relates to electromagnetic (EM) actuation of a scalar refractive field ψ and to devices, systems, and methods that generate, modulate, detect, and exploit ψ for propulsion, energy, communication, navigation/timing, metrology, shielding, computing, quantum control, and astrophysical environments.

Background

Early theoretical work, including Einstein (1911–12), considered the possibility that the speed of light might vary in a gravitational potential. Subsequent developments in modern scalar-field approaches introduce a refractive field ψ such that light propagates with optical index $n = e^{\psi}$ (phase velocity $v_{\mathrm{ph}} = c\,e^{-\psi}$) and matter accelerates according to

$\mathbf{a} = \frac{c^2}{2}\nabla\psi$. Applicant's prior filings address particular applications of this framework, including energy transmission, gravitational and signal correction, information encoding/computing, and artificial generation/control. However, the universal *actuation mechanism*—electromagnetic coupling to ψ in all regimes—has not yet been broadly protected. The present disclosure secures that umbrella.

Summary of the Invention

Disclosed are universal methods and systems wherein electromagnetic energy, in any configuration (static, dynamic, pulsed, resonant, nonresonant, stochastic, natural, or engineered), *actuates* a scalar refractive field ψ, producing measurable ψ perturbations that are subsequently used for: (i) propulsion/levitation, (ii) energy transmission/harvesting, (iii) communication and information processing (including longitudinal-capable channels and density-coded states), (iv) navigation and timing (route-dependent corrections), (v) metrology and sensing (cavities, clocks, interferometers), (vi) shielding/defense, (vii) quantum information control, and (viii) strong-field/astrophysical deployments. Extensions include dual-sector permittivity/permeability responses, anisotropy, nonlocal kernels, and stochastic ψ.

Brief Description of the Drawings

FIG. 1 shows an LC/cavity embodiment pumping ψ.
FIG. 2 shows a dual-sector $\epsilon(\psi)/\mu(\psi)$ split preserving $v_{\rm ph} = c/n$.
FIG. 3 shows propulsion via directed $\nabla\psi$.
FIG. 4 shows wireless energy via a refractive (ψ) corridor.
FIG. 5 shows ICC/computing with ψ modulation/detection.
FIG. 6 shows navigation/timing with vertically separated links and triangular loops.

Definitions and Notation

Throughout, ψ denotes a dimensionless scalar refractive field; $n = e^{\psi}$; $c_1 = c/n$ denotes one-way phase speed in a verified nondispersive band; matter acceleration is $\mathbf{a} = \frac{c^2}{2}\nabla\psi$. In a driven-mode picture, a ψ-mode amplitude q with natural frequency Ω_ψ obeys

$$\ddot{q} + 2\gamma_\psi \dot{q} + \Omega_\psi^2 q = (\lambda - 1)\, F_{\rm EM}(t),$$

where λ parameterizes EM back-reaction strength and $F_{\rm EM}(t)$ is a geometry-dependent EM functional. A dual-sector variant employs $\epsilon(\psi), \mu(\psi)$ with opposite response while preserving $v_{\rm ph} = 1/\sqrt{\epsilon\mu} = c/n$.

Detailed Description of Representative Embodiments

1. Universal EM$\rightarrow \psi$ Actuation

Any EM distribution—free-space fields, waveguides, cavities, coils, antennas, metamaterials, plasmas—couples into ψ. Actuation channels include: (i) direct drive at frequency ω, (ii) parametric pumping ($2\omega \approx \Omega_\psi$), (iii) DC bias with AC dither, (iv) impulsive/pulsed,

and (v) stochastic excitation. The coupling can be enhanced by maximizing the spatial overlap integral between EM energy density and a target ψ-mode support within a structure.

2. Laboratory Embodiments

LC/cavity: A driven LC tank or high-Q cavity accumulates field energy density, enhancing overlap with the ψ-mode. **Coils/Helmholtz:** DC/AC magnetic fields provide quasi-static or oscillatory actuation with scalable volumes. **Metamaterials:** Composite media engineer large local energy density and spatial gradients to maximize coupling.

3. Natural Embodiments

Ionospheric and magnetospheric structures, astrophysical plasmas, stellar interiors, accretion disks, and magnetospheres act as large-scale EM pumps of ψ, enabling passive or assisted actuation and measurement in situ.

4. Dual-Sector / Kernel / Stochastic Extensions

Dual-sector: permit $\epsilon(\psi) = \epsilon_0\, n\, e^{+\kappa\psi}$ and $\mu(\psi) = \mu_0\, n\, e^{-\kappa\psi}$ (linear or nonlinear) to encode sector-differential response while preserving $v_{\rm ph} = c/n$. **Anisotropy:** a bounded vector anisotropy $n_{ij} = e^{\psi}(\delta_{ij} + \alpha u_i u_j)$ yields direction-dependent couplings. **Nonlocal kernels:** ψ sourced via spatial convolution facilitates large-scale field shaping. **Stochastic ψ:** colored noise spectra imprint irreducible flicker in clocks/interferometers; controlled dithers enable system identification.

5. Applications

Propulsion/levitation: Directed $\nabla\psi$ produces effective thrust/inertial compensation. **Energy:** Wireless transmission via ψ-corridors to receivers that transduce density-field perturbations into electric/thermal/mechanical output. **Communications/ICC:** Information encoded in ψ modulations and longitudinal-capable channels; density-dependent interference enables logic and computing. **Navigation/timing:** Route-dependent corrections and environmental bias calibration using vertically separated links and triangular loops. **Metrology/sensing:** Cavity–atom ratios, atom/optical interferometry, accelerometry. **Shielding/defense:** Refractive/inertial shielding by shaping $n(\mathbf{x})$. **Quantum:** ψ-mediated control of phases and channels for state transfer and error suppression. **Astrophysical/strong-field:** Operation near compact objects (photon spheres/horizons) and in jets/plasmas.

Claims

Family A — Umbrella Universal Actuation

1. **(Independent, Method)** A method of actuating a scalar refractive field ψ comprising applying electromagnetic energy in any form, including static, dynamic, pulsed, resonant, nonresonant, parametric, or stochastic fields, such that the application produces a measurable perturbation of ψ.

2. The method of claim 1, wherein the electromagnetic energy is provided by at least one of: an LC circuit, a resonant cavity, a waveguide, an antenna, a coil, a capacitor, a metamaterial, a plasma, or a laser.

3. The method of claim 1, wherein the actuation comprises parametric pumping with a drive frequency approximately twice a ψ-mode frequency.

4. **(Independent, Device/System)** A device configured to carry out the method of claim 1, the device comprising at least one electromagnetic source and a structure arranged to maximize coupling to ψ.

Family B — Dual-Sector, Anisotropy, Kernels, Stochastic

5. The method of claim 1, wherein a dual-sector response is implemented with $\epsilon(\psi)$ and $\mu(\psi)$ that vary oppositely while preserving $v_{\mathrm{ph}} = c/n$.

6. The method of claim 1, wherein coupling employs at least one of: anisotropic response, nonlocal convolution kernels, or stochastic excitation of ψ.

7. The device of claim 4, wherein the structure comprises metamaterials configured to enhance overlap between electromagnetic energy density and a ψ-mode support.

Family C — Propulsion / Levitation

8. A method of producing propulsion or levitation comprising actuating ψ according to claim 1 to generate a directed gradient $\nabla\psi$ that imparts effective acceleration to a platform.

9. The method of claim 8, further comprising shaping a thrust vector subject to safety constraints on at least one of field magnitude, gradient magnitude, spatial extent, and temporal slew rate.

10. The device of claim 4 configured with control electronics operative to synthesize $\nabla\psi$ profiles that produce thrust or inertial compensation on a platform.

Family D — Energy / Wireless Power

11. A method of wireless energy transmission comprising actuating ψ at a transmitter to form a refractive corridor, propagating energy along the corridor, and receiving said energy at a receiver that transduces ψ perturbations into at least one of electrical, thermal, or mechanical output.

12. The method of claim 11, wherein the receiver comprises an interferometric or density-sensitive detector tuned to ψ modulations.

13. The device of claim 4 further comprising a transmitter configured to impose ψ-corridor modulation and a receiver configured to convert received ψ perturbations to usable power.

Family E — Communications / ICC / Computing

14. A method of communication comprising encoding information into ψ modulations actuated per claim 1 and transmitting along a channel supporting longitudinal-capable modes, and decoding at a receiver by measuring local variations of $n = e^{\psi}$.

15. A computing architecture comprising: a ψ modulator; a propagation channel; and a detector that performs logic operations using interference of ψ-encoded states.

16. A non-transitory computer-readable medium storing instructions that, when executed by control electronics of the device of claim 4, cause electromagnetic sources to actuate ψ according to an information-encoding protocol and to demodulate received ψ signals.

Family F — Navigation / GPS / Timing

17. A method of navigation or timing comprising actuating and/or measuring ψ along at least two routes of differing altitude or geometry to compute route-dependent corrections and environmental bias calibrations.

18. The method of claim 17, wherein the routes include vertically separated fiber links and triangular loops among three stations with active path swapping to isolate nonreciprocal delays attributable to ψ.

Family G — Metrology / Sensing

19. A method of metrology comprising actuating ψ and measuring resulting perturbations using at least one of: optical cavities, atomic clocks, optical or matter-wave interferometers, accelerometers, or frequency-comb timing links.

20. The method of claim 19, wherein a ratio of cavity-stabilized to atomic frequency exhibits a non-null slope versus gravitational potential difference due to sector-differential response in a verified nondispersive band.

Family H — Shielding / Defense

21. A method of shielding comprising shaping $n(\mathbf{x}) = e^{\psi(\mathbf{x})}$ to deflect, attenuate, or mitigate incident radiation or impacts via refractive steering or inertial compensation.

Family I — Quantum Information

22. A method of quantum control comprising actuating ψ to engineer phase evolution, channel properties, or error suppression in quantum state transfer or entanglement distribution.

Family J — Strong-Field / Astrophysical

23. A method of operating near compact objects comprising actuating or measuring ψ in environments including photon spheres, horizons, jets, or magnetospheres.

24. The method of claim 23, wherein actuation leverages natural plasma or magnetospheric electromagnetic fields to pump ψ.

Figures

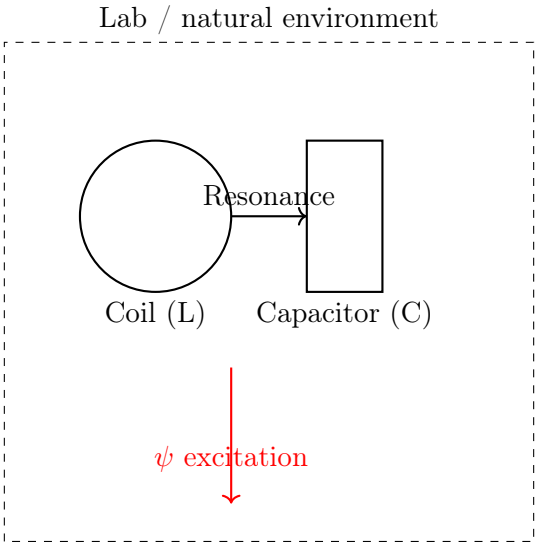

Figure 1: FIG. 1 — Electromagnetic resonance (LC/cavity) inducing a ψ-mode.

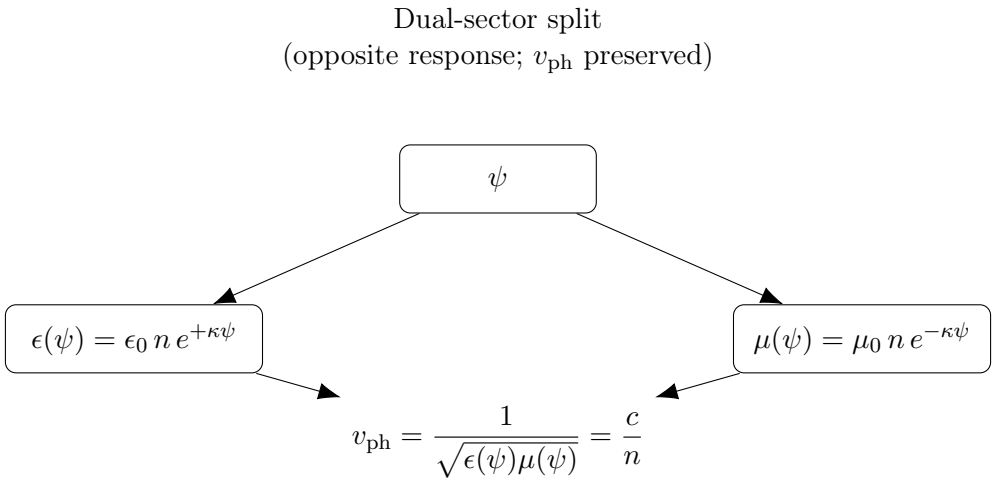

Figure 2: FIG. 2 — Dual-sector ϵ/μ split preserving $v_{\mathrm{ph}} = c/n$ while enabling sector-differential response.

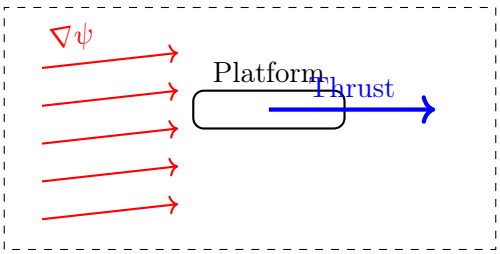

Figure 3: FIG. 3 — Propulsion/levitation via directed $\nabla\psi$.

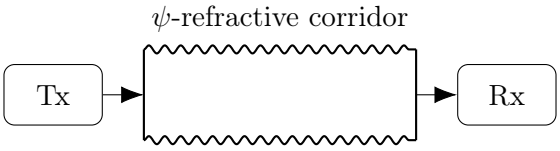

Figure 4: FIG. 4 — Wireless energy/signaling via a ψ-corridor to a density-sensitive receiver.

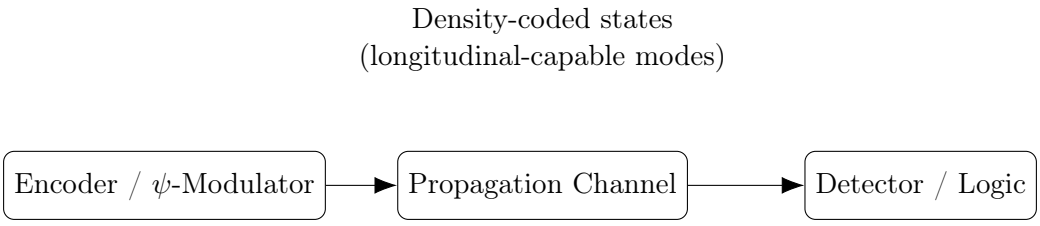

Figure 5: FIG. 5 — ICC/computing with ψ-encoded information and density-dependent interference.

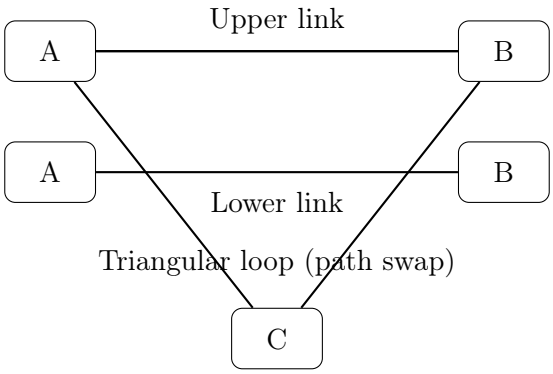

Figure 6: FIG. 6 — Navigation/timing: vertically separated links and triangular loop isolate route-dependent ψ effects.